

Data Poisoning Detection Pipeline

A reproducible CIFAR-10 prototype for recovering from label flips and mitigating backdoors

Two attack-specific solutions: Cleanlab label recovery and known-trigger backdoor mitigation

Repository: github.com/VIDITJAIN7/Data-Poisoning-Detection-Pipeline-for-SCD-2026

+1.32 pp

Label-flip accuracy recovered

-94.11 pp

Backdoor attack success rate reduced

89.52%

Normal accuracy retained

Threat	Selected solution	Headline result
Label flips	Cleanlab issue removal	87.33% to 88.65% accuracy
Backdoor patch	Known-trigger mitigation	95.33% to 1.22% ASR at 89.52% accuracy

Abstract

We present a reproducible CIFAR-10 pipeline for detecting and mitigating training-data poisoning before model deployment. Two defenses are evaluated independently: Cleanlab issue removal for targeted label flips, and known-trigger filtering for BadNets-style backdoors. On held-out test data, label-flip accuracy recovers from 87.33% to 88.65%, and backdoor attack success rate falls from 95.33% to 1.22% while 89.52% normal accuracy is retained.

1	2	3	4	5
Project objective	Proposed solution	Solution validation	Results	Conclusions

1 Project objective

Training data can be compromised before model training. The project addresses two practical failure modes in image classification: **targeted label flips** that teach the wrong class, and **backdoor triggers** that cause a hidden target prediction when a patch is present.

Business problem. An ML team needs a repeatable pre-deployment check that can detect suspicious training data, recover useful labels, and verify that a model no longer responds to a trigger.

2 Proposed solution



Defenses observe images and labels only

Poison membership stays evaluation-only

Layer	Implementation	Why it matters
Data integrity	Stable CIFAR-10 IDs, fixed split, isolated evaluation metadata	Prevents leakage and makes runs auditable
Attack construction	Airplane-to-truck flips and a 3x3 white bottom-right patch	Reproduces benchmark scenarios
Detection	Cleanlab 3-fold OOF probabilities with activation and spectral baselines	Scores samples without self-training bias
Correction	Remove selected label issues; fine-tune or prune the backdoor model	Preserves unselected observed data
Evaluation	Normal accuracy, per-class accuracy, ASR, triggered accuracy	Separates utility from security

Deployment fit

- Designed for research or MLOps teams protecting training pipelines before production release.
- Uses established tooling and benchmark algorithms instead of an unvalidated custom defense.
- Replaces ad hoc retraining with fixed splits, manifests, checkpoints, and measurable acceptance criteria.

Innovation. The layered workflow itself: trusted data, attack-aware validation, and separate utility and security reporting in one reproducible prototype.

3 Solution validation

The active campaign uses ResNet-18 with fixed seeds over a stratified CIFAR-10 split. The untouched 10,000-image test set is reserved for final evaluation only.

ResNet-18	30	45,000	2,000	3,000	10,000
Architecture	Training epochs	Attack pool	Trusted images	Development set	Held-out test

Validation checks

- Stable IDs and split disjointness are checked before attack construction.
- Label-flip count and backdoor patch rendering are verified at the dataset boundary.
- OOF folds exclude each held-out sample from fitting, reducing self-training bias.
- Corrected datasets retain every unselected observed image and label.
- Backdoor ASR is evaluated on triggered non-target test images, separately from normal accuracy.
- Runs write manifests, metrics, checkpoints, and readable progress logs.

Evidence and limitations

Dimension	Measured evidence	Interpretation
Reproducibility	Fixed seeds, manifests, campaign directories	Scenario can be rerun from the README
Label defense	Cleanlab selection; 5% label-flip results	Useful recovery, not complete restoration
Backdoor defense	ASR measured before and after mitigation	Security improvement is measured directly
Limitation	Selected backdoor result assumes the fixed trigger is known	Not evidence of general unknown-trigger detection

Correctness principle. High clean accuracy alone does not prove backdoor removal; ASR is the security metric.

Reproduction

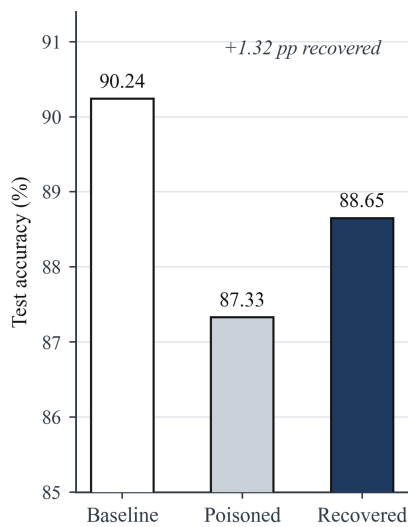
```
pip install -r requirements.txt
bash scripts/run_layered_campaign.sh
python scripts/generate_results_summary.py
```

Active artifacts under `outputs/layered_campaign/` and `outputs/results/`; archived records under `outputs/archive/testing/`.

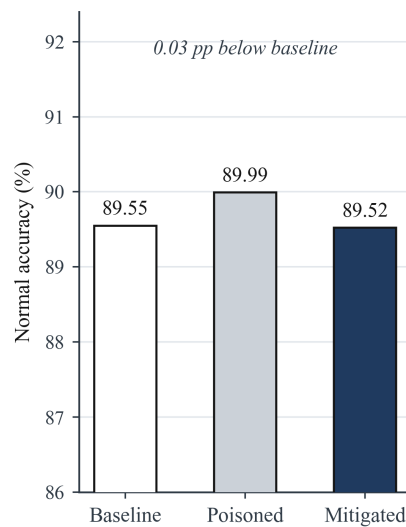
4 Results

The following are the strongest completed results selected for the two attack-specific solutions. Percentages are test-set measurements from the recorded runs.

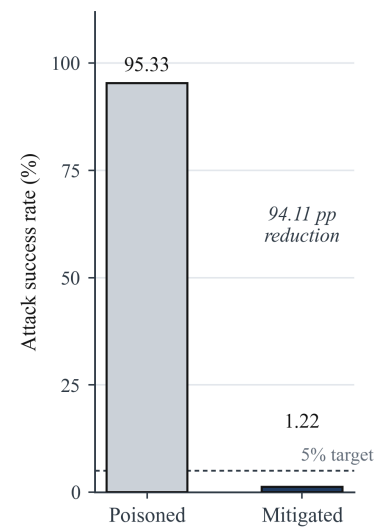
Solution	Baseline	Poisoned	After solution	Improvement
Label-flip recovery	90.24%	87.33%	88.65%	+1.32 pp
Backdoor mitigation, accuracy	89.55%	89.99%	89.52%	-0.47 pp vs baseline
Backdoor mitigation, ASR	95.33%	95.33%	1.22%	-94.11 pp



(a) Label-flip recovery



(b) Utility retained



(c) Backdoor neutralised

Interpretation

The label solution improves the poisoned model by **1.32 percentage points**, but its selected run remains below its baseline. The backdoor solution is selected for **security**: it reduces attack success from **95.33% to 1.22%** while retaining **89.52%** normal accuracy, only 0.03 percentage points below baseline. The two solutions target different attack mechanisms and are not presented as one universal checkpoint.

Reading the two solutions together. Utility and security are reported independently, so a gain in one is never used to imply a gain in the other.

5 Conclusions

What we accomplished

- Delivered a reproducible end-to-end CIFAR-10 pipeline covering attack injection, detection, correction, retraining, and evaluation.
- Implemented a controlled stratified split with 45,000 attack-pool images, 2,000 trusted images, 3,000 development images, and a held-out 10,000-image test set.
- Recovered label-flip accuracy from 87.33% to 88.65% using Cleanlab issue removal (+1.32 percentage points over the poisoned model).
- Reduced backdoor attack success rate from 95.33% to 1.22% while retaining 89.52% normal accuracy.
- Reported utility and security independently through normal accuracy and attack success rate rather than treating accuracy alone as proof of safety.
- Preserved stable sample IDs, run manifests, JSON logs, campaign outputs, and archived testing records for audit and rerun.
- Packaged reproduction as installable code with a single layered campaign runner and summary generation scripts.

Final conclusion

The prototype shows a credible recovery path for label flips and a strong fixed-trigger backdoor mitigation result, with measurable trade-offs instead of unsupported guarantees.

+1.32 pp

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-94.11 pp

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Sources

- Cleanlab documentation on out-of-sample probabilities.
- Trusted AI Adversarial Robustness Toolbox documentation.
- Tran, Li and Madry. *Spectral Signatures in Backdoor Attacks*. NeurIPS 2018.
- Chen et al. *Detecting Backdoor Attacks on Deep Neural Networks by Activation Clustering*. 2019.
- Gu, Dolan-Gavitt and Garg. *BadNets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain*. 2017.